

Portfolio Construction: Where Markowitz Falls Short

® Abstract: This radar series is intended to provide a deep dive into portfolio construction. As the first instalment, this Weekly Quant Radar compares the naïve equal-weighted $1/n$ portfolio with the classical Markowitz mean-variance optimiser. While mean-variance is theoretically superior, its practical performance is notoriously brittle. We show that with realistic estimation error - even for moderate asset universes (e.g., 40-50 assets) - Markowitz often fails to outperform the simple $1/n$ rule out-of-sample. The discussion explains why estimation error in expected returns and covariances undermines the optimiser and discusses conditions under which $1/n$ remains a competitive benchmark, highlighting the fundamental trade-off between theoretical optimality and robustness in active portfolio construction.

An intuitive starting point to portfolio construction is the equal-weighted, or $1/n$, rule. An investor allocates wealth across n assets by assigning the same weight to each:

$$w_i = \frac{1}{n}, \quad i = 1, \dots, n.$$

where w_i is the share of wealth in asset i . The portfolio satisfies $\sum_{i=1}^n w_i = 1$. Let R_i denote the return on asset i . The portfolio return is then given by:

$$R_p = \sum_{i=1}^n w_i R_i = \frac{1}{n} \sum_{i=1}^n R_i.$$

The expected portfolio return is $\mathbb{E}[R_p] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[R_i]$. Portfolio variance depends on covariances:

$$\text{Var}(R_p) = w^\top \Sigma w = \left(\frac{1}{n} \mathbf{1} \right)^\top \Sigma \left(\frac{1}{n} \mathbf{1} \right).$$

Now consider the mean-variance framework of Markowitz (1952), a more involved and active alternative to the $1/n$ allocation rule. Its goal is to choose weights optimally by trading off expected return against risk. Let μ denote the vector of expected asset returns. The expected portfolio return is $\mathbb{E}[R_p] = w^\top \mu$. The investor chooses w to minimise variance for a given target return μ_p :

$$\min_w w^\top \Sigma w \quad \text{subject to} \quad w^\top \mu = \mu_p, \quad \sum_{i=1}^n w_i = 1.$$

The first constraint fixes the desired expected return, the second ensures full investment. Solving for different μ_p yields the efficient frontier: the set of portfolios offering the highest expected return for each risk level.

In application, however, μ and Σ are unknown and must be estimated from historical data (sample means and covariances). Estimation error - especially in μ - leads to optimised weights that are extreme, unstable and fit noise rather than signal. Even with a moderate universe of 40-50 assets, the number of parameters grows quadratically. With typical sample sizes (e.g., 5-10 years of monthly data), the signal-to-noise ratio is so unfavourable that the out-of-sample performance of the Markowitz portfolio often falls short of the equal-weighted benchmark.

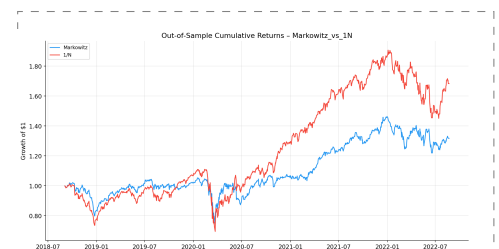
Since the $1/n$ rule requires no estimation and guarantees diversification, it has proven to be remarkably difficult to beat systematically out-of-sample. This phenomenon has been explored in the seminal work by DeMiguel, Garlappi & Uppal (2009), which shows that no optimised actively managed strategy consistently dominates the $1/n$ rule out-of-sample. The reasons are twofold: (i) expected returns are extremely difficult to estimate precisely, and (ii) the optimiser aggressively magnifies estimation errors by taking large long/short positions in assets with apparently high Sharpe ratios. The equal-weighted portfolio, by ignoring return estimates entirely, avoids this overfitting.

The fragility of active portfolio management through optimisation has deeper roots. Michaud (1989) famously characterised mean-variance optimisation as an “error maximizer,” noting that optimal portfolios are “financially meaningless” when inputs are noisy. This critique has been widely accepted, and new methodologies have emerged in response. The challenge for any active approach is therefore not theoretical optimality, but overcoming estimation error to deliver stable out-of-sample performance relative to the simple and robust $1/n$ benchmark.

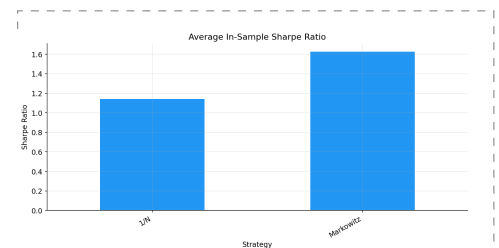
(The next instalment will discuss robust mean-variance variants - shrinkage, Bayesian priors and weight constraints - and when they can realistically surpass the simple $1/n$ rule.)

Charts

Out-of-sample Returns



In-sample Sharpe Ratio



Source: prometheuscapi.nl

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